

Irina Gaynanova

irinagain.github.io

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Research interests

Statistical/machine learning, data integration, high-dimensional biomedical data, multivariate analysis, computational statistics, applications to multi-omics data and data from wearable devices (continuous glucose monitors, ambulatory blood pressure monitors, activity trackers), cardiometabolic health.

Education

- Ph.D., Statistics 05/2015
Cornell University, Ithaca, NY
Advisors: James Booth and Martin Wells
- M.S., Statistics 05/2013
Cornell University, Ithaca, NY
- Diploma with honors (M.S.), Applied Mathematics and Computer Science 06/2009
Lomonosov Moscow State University, Moscow, Russia

Professional Positions

- *Associate Professor (by courtesy)* since 01/2025
Department of Statistics, University of Michigan
- *Associate Professor (with tenure)* since 09/2023
Department of Biostatistics, University of Michigan
- *Adjunct Associate Professor* since 09/2023
Department of Statistics, Texas A&M University
- *Associate Professor (with tenure)* 09/2021 - 08/2023
Department of Statistics, Texas A&M University
- *Assistant Professor* 07/2015 - 08/2021
Department of Statistics, Texas A&M University
- *Senior Specialist* 06/2009 - 07/2010
Balancing Market Division, OJSC Trading System Administrator, Moscow, Russia
- *Junior Statistician* 06/2008 - 05/2009
Census Division, AC Nielsen, Moscow, Russia

Awards and Honors

- University of Michigan Honored Instructor, Michigan Housing 2025
- COPSS Emerging Leader Award 2025
- IMS Thelma and Marvin Zelen Emerging Women Leaders in Data Science Award 2025
- Elected member, International Statistical Institute 2024
- NSF CAREER Award 2021
- Dr. Judith Edmiston Mentoring Award, College of Science, Texas A&M University 2021
- DeBakey Executive Research Leadership Certification, Texas A&M University 2021
- David P. Byar Young Investigator Award, ASA Biometrics Section 2018
- Cornelia Ye Outstanding Teaching Assistant Award, Cornell University 2014
- EducationUSA Opportunity Award 2009
- Study Abroad Scholarship, Technical University of Munich, Germany 2007

Competitive Travel Awards

- IMS New Researchers Conference 2017
- NISS Writing Workshop for Junior Researchers 2016
- SAMSI LDHD Summer School Program 2013
- Building Future Faculty Program, NC State University 2013
- Diversity and Mentoring Program at JSM 2012

Publications and Submitted Manuscripts

Note on authorship: the sign * at the beginning of a paper indicates alphabetical order of authorships; the sign + indicates co-first authorships; underline denotes mentored student or post-doctoral associate; the sign ☒ denotes the corresponding author.

Submitted/Under Revision Manuscripts:

59. Park J. and **Gaynanova I.**☒(2026+) "Fréchet regression of multivariate distributions with nonparanormal transport." Preprint available on [arXiv:2603.07014](https://arxiv.org/abs/2603.07014).
58. Shao E., Park J., Punjabi N., Jiang H. and **Gaynanova I.**☒(2026+) "Fast distance computation of multivariate distributions via nonparanormal transport." Preprint available on [arXiv:2603.00322](https://arxiv.org/abs/2603.00322).
57. Qian L., Aurora R.N, Punjabi N. and **Gaynanova I.**☒(2026+) "Smooth tensor decomposition with application to ambulatory blood pressure monitoring data." Preprint available on [arXiv:2507.1172](https://arxiv.org/abs/2507.1172).
56. Kim R. and **Gaynanova I.**☒(2026+) "A sparse linear model for positive definite estimation of covariance matrices." Preprint available on [arXiv:2503.09026](https://arxiv.org/abs/2503.09026).
55. James C., Yuan D., **Gaynanova I.** and Arroyo J.☒(2026+) "Learning Joint and Individual Structure in Network Data with Covariates." Preprint available on [arXiv:2406.08776](https://arxiv.org/abs/2406.08776).
54. Yuan D., Zhang Y., Guo S., Wang W. and **Gaynanova I.**☒(2026+) "Exponential canonical correlation analysis with orthogonal variation." Preprint available on [arXiv:2208.00048](https://arxiv.org/abs/2208.00048).

Peer-reviewed Publications:

53. Xu C., Schmidt B.M., Krambrink A., Park J., Song P., Jones T., Holmes C., Chen K., Ye W., Kolenic G., Roy S., Sen C., Pop-Busui R., Spino C. and **Gaynanova I.** (2026) "Longitudinal Healing and Amputation Trajectories in Diabetic Foot Ulcers: Predictive Power of Wound Area and Duration and Sample-Size Implications from the Diabetic Foot Consortium." *Wound Repair and Regeneration*, Vol. 34, No. 4, accepted.
Link: <https://doi.org/10.1111/wrr.70189>
52. Choi J.[✉], Chung H.C., **Gaynanova I.** and Ni Y. (2026+) "Bayesian segmented Gaussian copula factor model for single-cell sequencing data." *Bayesian Analysis*, advance publication.
Link: <https://doi.org/10.1214/25-BA1559>
51. Williamson W., Kok N., Lee J., Wiens J. and **Gaynanova I.** (2026) "Social Determinants of Health and Continuous Glucose Monitoring Metrics in Type 1 or Type 2 Diabetes." *JAMA Network Open*, Vol. 9, No. 7, e2621290.
Link: <https://doi.org/10.1001/jamanetworkopen.2026.21290>
50. Park J., Kok N. and **Gaynanova I.** (2026) "Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions." *Statistics in Medicine*, Vol. 45, No. 15-17, e70646.
Link: <https://doi.org/10.1002/sim.70646>
49. Kok N., Williamson W., Lee J. and **Gaynanova I.** (2026) "Impact of Missing Data and Monitoring Duration on Downstream Analyses in Continuous Glucose Monitoring." *Diabetes Care*, Vol. 49, No. 6, 1031-1039.
Link: <https://doi.org/10.2337/dc25-2935>
48. Girgis C.[✉], **Gaynanova I.**, Behme S., Herman W.H., Holmes, C., Schmidt B.M. (2026+) "Do Immediate Perioperative Glucose Measurements Predict Outcomes in Non-Elective Pedal Amputation?" *Foot & Ankle Specialist*, advance publication.
Link: <https://doi.org/10.1177/19386400261420316>
47. Sergazinov R., Taeb A. and **Gaynanova I.** (2026) "A spectral method for multi-view subspace learning using the product of projections." *Biometrika*, Vol. 113, No. 2, asaf088.
Link: <https://doi.org/10.1093/biomet/asaf088>
46. Wang L., Ni Y. and **Gaynanova I.** (2026) "Truncated Gaussian copula principal component analysis with application to pediatric acute lymphoblastic leukemia patients' gut microbiome." *Statistical Methods in Medical Research*, Vol. 35, No. 3, 443-455.
Link: <https://doi.org/10.1177/09622802251412844>
45. Williamson W., Lee J.M. and **Gaynanova I.** (2026) "A processing algorithm to address real-world data quality issues with continuous glucose monitoring data." *Journal of Diabetes Science and Technology*, Vol. 20, No. 4, 1376-1380.
Link: <https://doi.org/10.1177/19322968251319801>
44. Chung H.C., Ni Y. and **Gaynanova I.** (2025) "Sparse semiparametric discriminant analysis for high-dimensional zero-inflated data." *Journal of Machine Learning Research*, Vol. 26, No. 210, 1-54.
Link: <https://jmlr.org/papers/v26/24-0046.html>
43. Coulter A.[✉], Lee R. and **Gaynanova I.** (2025) "fastfrechet: an R package for fast implementation of Frechet regression with distributional responses." *The Journal of Open Source Software*, Vol. 10, No. 109, 7925.
Link: <https://doi.org/10.21105/joss.07925>

42. Coulter A., Aurora R. N., Punjabi N. and **Gaynanova I.** (2025) "Fast variable selection for distributional regression with application to continuous glucose monitoring data." *Annals of Applied Statistics*, Vol. 19, No. 3, 2105-2128.
Link: <http://dx.doi.org/10.1214/25-AOAS2038>
41. **Gaynanova I.** and Lee J. M. (2025) "When algorithms diverge: quantification of glycemic episodes from CGM data." *Diabetes Technology & Therapeutics*, Vol. 27, No. 6, 500-502.
Link: <https://doi.org/10.1089/dia.2024.0618>
40. Chun E., Fernandes N. and **Gaynanova I.** (2024) "An update on the iglu software package for interpreting continuous glucose monitoring data." *Diabetes Technology & Therapeutics*, Vol. 26, No. 12, 939-950.
Link: <https://doi.org/10.1089/dia.2024.0154>
39. Wang Z., **Gaynanova I.**, Aravkin A. and Risk B. (2024) "Sparse Independent Component Analysis with an Application to Cortical Surface fMRI Data in Autism." *Journal of the American Statistical Association*, Vol. 119, No. 548, 2508-2520.
Link: <https://doi.org/10.1080/01621459.2024.2370593>
[Paper won JASA 2024 Reproducibility Award.]
38. Wang L., **Gaynanova I.** and Risk B. (2024) "singR: An R Package for Simultaneous Non-Gaussian Component Analysis for Data Integration" *The R Journal*, Vol. 15, No. 4, 69-83.
Link: <https://doi.org/10.32614/RJ-2023-084>
37. **Gaynanova I.** and Sergazinov R. (2024) "Comments on "Data Integration Via Analysis of Subspaces (DIVAS)." *TEST*, Vol 33, 675-682.
Link: <https://doi.org/10.1007/s11749-024-00936-8>
36. Sergazinov R., Chun E., Rogovchenko V., Fernandes N., Kasman N. and **Gaynanova I.** (2024) "GlucoBench: Curated List of Continuous Glucose Monitoring datasets with Prediction Benchmarks." *International Conference on Learning Representations (ICLR)*.
Link: <https://openreview.net/forum?id=cUSNs8nGaV>
35. Whitfield-Cargile C.M., Chung H.C., Coleman M.C., Cohen N.D., Chamoun-Emanuelli A.M., Ivanov I., Goldsby J.S., Davidson L.A, **Gaynanova I.**, Ni Y. and Chapkin R.S. (2024) "Integrated analysis of gut metabolome, microbiome, and exfoliome data in an equine model of intestinal injury." *Microbiome*, Vol. 12, No. 74
Link: <https://doi.org/10.1186/s40168-024-01785-1>
34. Chun E., **Gaynanova I.**, Melanson E. and Lyden, K. (2024) "Pre- Versus Postmeal Sedentary Duration; Impact on Postprandial Glucose in Older Adults with Overweight or Obesity." *Journal for the Measurement of Physical Behaviour*, Vol. 7, No. 1.
Link: <https://doi.org/10.1123/jmpb.2023-0032>
33. Cheong, SM and **Gaynanova I.** (2024) "Sensing the impact of extreme heat on physical activity and sleep." *Digital Health*, 10.
Link: <https://doi.org/10.1177/20552076241241509>
32. Yi S., Wong R. and **Gaynanova I.** (2023) "Hierarchical nuclear norm penalization for multi-view data integration." *Biometrics*, Vol. 79, No. 4, 2933-2946.
Link: <http://doi.org/10.1111/biom.13893>
31. Sergazinov R., Leroux A., Cui E., Crainiceanu C., Aurora R., Punjabi N. and **Gaynanova I.** (2023) "A case study of glucose levels during sleep using fast function on scalar regression inference." *Biometrics*, Vol. 79,

- No. 4, 3873-3882.
Link: <https://doi.org/10.1111/biom.13878>
30. Sergazinov R.[✉], Armandpour R. and **Gaynanova I.** (2023) "Gluformer: Transformer-based personalized glucose forecasting with uncertainty quantification." *2023 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.
Link: <https://doi.org/10.1109/ICASSP49357.2023.10096419>
 29. Zhang Y. and **Gaynanova I.**[✉] (2022) "Joint association and classification analysis of multi-view data." *Biometrics*, Vol. 78, No. 4, 1614-1625.
Link: <https://doi.org/10.1111/biom.13536>
[Paper won SLDS (Statistical Learning and Data Science) student paper award.]
 28. Yuan D. and **Gaynanova I.**[✉] (2022) "Double-matched matrix decomposition for multi-view data." *Journal of Computational and Graphical Statistics*, Vol. 31, No. 4, 1114-1126.
Link: <https://doi.org/10.1080/10618600.2022.2067860>
 27. Chung H.C., **Gaynanova I.** and Ni Y.[✉] (2022) "Phylogenetically informed Bayesian truncated copula graphical models for microbial association networks." *Annals of Applied Statistics*, Vol. 16, No. 4, 2437-2457.
Link: <https://doi.org/10.1214/21-AOAS1598>
 26. Schwenck J.[✉], Punjabi N. and **Gaynanova I.** (2022) "bp: Blood Pressure analysis in R." *PLoS One*, Vol. 17, No. 9, e0268934.
Link: <https://doi.org/10.1371/journal.pone.0268934>
 25. Aurora R. N.[✉], **Gaynanova I.**, Patel P., and Punjabi N. M. (2022). "Glucose profiles in obstructive sleep apnea and type 2 diabetes mellitus." *Sleep Medicine*, Vol. 95, 105-111
Link: <https://doi.org/10.1016/j.sleep.2022.04.007>
 24. Fernandes N., Nhan, N., Chun, E., Punjabi N. and **Gaynanova I.**[✉] (2022) "Open-source algorithm to calculate mean amplitude of glycemic excursions using short and long moving averages." *Journal of Diabetes Science and Technology*, Vol. 16, No. 2, 576-577.
Link: <https://doi.org/10.1177/19322968211061165>
 23. **Gaynanova I.**[✉], Punjabi N. and Crainiceanu C. (2022) "Modeling continuous glucose monitoring (CGM) data during sleep." *Biostatistics*, Vol. 23, No. 1, 223-239.
Link: <https://doi.org/10.1093/biostatistics/kxaa023>
 22. Lapanowski A. and **Gaynanova I.**[✉] (2021) "Compressing large-sample data for discriminant analysis." *2021 IEEE International Conference on Big Data (Big Data)*, 1068-1076.
Link: <https://doi.org/10.1109/BigData52589.2021.9671676>
 21. Huang M., Müller C. and **Gaynanova I.** (2021) "latentcor: An R package for estimating latent correlations from mixed data types." *Journal of Open Source Software*, Vol. 6, No. 65, 3634.
Link: <https://doi.org/10.21105/joss.03634>
 20. Risk B.[✉] and **Gaynanova I.**⁺ (2021) "Simultaneous non-Gaussian component analysis (SING) for data integration in neuroimaging." *Annals of Applied Statistics*, Vol. 15, No. 3, 1431-1454.
Link: <https://doi.org/10.1214/21-AOAS1466>
 19. Yoon G., Müller C. and **Gaynanova I.**[✉] (2021) "Fast computation of latent correlations." *Journal of Computational and Graphical Statistics*, Vol. 30, No. 4, 1249-1256.
Link: <https://doi.org/10.1080/10618600.2021.1882468>

18. Solhjoo S., Punjabi N., Ivanescu A., Crainiceanu C., **Gaynanova I.**, Wicken C., Buckenmaier C., and Haigney M. (2021) "Methadone destabilizes cardiac repolarization during sleep." *Clinical Pharmacology & Therapeutics*, Vol. 110, No. 4, 1066-1074.
Link: <https://doi.org/10.1002/cpt.2368>
17. Broll S., Urbanek J., Buchanan D., Chun E., Muschelli J., Punjabi N. and **Gaynanova I.** (2021) "Interpreting blood glucose data with R package iglu." *PLoS one*, Vol. 16, No. 4, e0248560.
Link: <https://doi.org/10.1371/journal.pone.0248560>
16. Taylor N., **Gaynanova I.**, Eschrich S., Welsh E., Garrett T., Beecher C., Sharma R., Koomen J., Smalley K., Messina J., Kanetsky P. (2020) "Metabolomics of primary cutaneous melanoma and matched adjacent extratumoral microenvironment." *PLoS one*, 15(10):e0240849.
Link: <https://doi.org/10.1371/journal.pone.0240849>
15. Yoon G., Carroll R. and **Gaynanova I.** (2020) "Sparse semiparametric canonical correlation analysis for data of mixed types." *Biometrika*, Vol. 107, No. 3, 609-625
Link: <https://doi.org/10.1093/biomet/asaa007>
[ENAR best poster award.]
14. **Gaynanova I.** (2020) "Prediction and estimation consistency of sparse multi-class penalized optimal scoring." *Bernoulli*, Vol. 26, No. 1, 286-322.
Link: <https://projecteuclid.org/euclid.bj/1574758829>
13. **Gaynanova I.** and Li G. (2019). "Structural learning and integrative decomposition of multi-view data." *Biometrics*, Vol. 75, No. 4, 1121-1132.
Link: <https://doi.org/10.1111/biom.13108>
[Paper selected for 2018 David P. Byar Young Investigator Award.]
12. * Bien J., **Gaynanova I.**, Müller C. and Lederer J. (2019). "Prediction error bounds for linear regression with the TREX." *TEST*, Vol. 28, No. 2, 451-474.
Link: <https://doi.org/10.1007/s11749-018-0584-4>
11. Yoon G., **Gaynanova I.** and Müller, C. (2019) "Microbial networks in SPRING - Semi-parametric rank-based correlation and partial correlation estimation for quantitative microbiome data." *Frontiers in Genetics*, Vol. 10, 516.
Link: <https://doi.org/10.3389/fgene.2019.00516>
10. Lapanowski A. and **Gaynanova I.** (2019) "Sparse feature selection in kernel discriminant analysis via optimal scoring." *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 89, 1704-1713.
Link: <http://proceedings.mlr.press/v89/lapanowski19a.html>
9. Lederer J., Lu Y. and **Gaynanova I.** (2019). "Oracle inequalities for high-dimensional prediction." *Bernoulli*, Vol. 25, No. 2, 1225-1255.
Link: <https://projecteuclid.org/euclid.bj/1551862849>
8. **Gaynanova I.** and Wang T. (2019) "Sparse quadratic classification rules via linear dimension reduction." *Journal of Multivariate Analysis*, Vol. 169, 278-299.
Link: <https://doi.org/10.1016/j.jmva.2018.09.011>
7. **Gaynanova I.**, Urbanek J. and Punjabi N. (2018). "Letter to the Editor: Corrections of equations on glycemic variability and quality of glycemic control." *Diabetes Technology & Therapeutics*, Vol. 20, No. 4, 317.
Link: <https://doi.org/10.1089/dia.2018.0057>

6. Li G.✉ and **Gaynanova I.** (2018). "A general framework for association analysis of heterogeneous data." *Annals of Applied Statistics*, Vol. 12, No. 3, 1700-1726.
Link: <http://doi.org/10.1214/17-AOAS1127>
5. * Bien J., **Gaynanova I.**✉, Müller C. and Lederer J. (2018). "Non-convex global minimization and false discovery rate control for the TREX." *Journal of Computational and Graphical Statistics*, Vol. 27, No. 1, 23-33.
Link: <https://doi.org/10.1080/10618600.2017.1341414>
4. Hokamp J.✉, Leidy S., **Gaynanova I.**, Cianciolo R. and Nabity M. (2018) "Correlation of electrophoretic urine protein banding patterns with severity of renal damage in dogs with proteinuric chronic kidney disease." *Veterinary Clinical Pathology*, Vol. 47, No. 3, 424-434.
Link: <https://doi.org/10.1111/vcp.12648>
3. **Gaynanova I.**✉, Booth J. and Wells M. (2017). "Penalized versus constrained generalized eigenvalue problems." *Journal of Computational and Graphical Statistics*, Vol. 26, No. 2, 379-387.
Link: <https://doi.org/10.1080/10618600.2016.1172017>
2. **Gaynanova I.**✉, Booth J. and Wells M. (2016). "Simultaneous sparse estimation of canonical vectors in the $p \gg n$ setting." *Journal of the American Statistical Association*, Vol. 111, No. 514, 696-706.
Link: <http://doi.org/10.1080/01621459.2015.1034318>
1. **Gaynanova I.**✉ and Kolar M. (2015). "Optimal variable selection in multi-group sparse discriminant analysis." *Electronic Journal of Statistics*, Vol. 9, No. 2, 2007-2034.
Link: <http://doi.org/10.1214/15-EJS1064>

Unrefereed Manuscripts and Other Written Products:

4. **Gaynanova I.** (2024) Public comment to FDA in response to "CDRH seeks public comment: Digital Health Technologies for Detecting Prediabetes and Undiagnosed Type 2 Diabetes." Docket No. FDA-2023-N-4853.
3. **Gaynanova I.**✉ (2022) "Digital biomarkers of glucose control - reproducibility challenges and opportunities." *Biopharmaceutical Report*, Vol. 29, No. 1, 21-26.
2. **Gaynanova I.** (2015). "Estimation of sparse low-dimensional linear projections." Ph.D. Thesis, Cornell University.
1. **Gaynanova I.**, Booth J. and Wells M. (2013). "Supervised classification using sparse Fisher's LDA." Technical report, [arXiv:1301.4976 \[stat.ML\]](https://arxiv.org/abs/1301.4976).

Software

Mentored student and post-doctoral associate co-authors are underlined.

18. **iglu**: an R package for interpreting data from continuous glucose monitors (CGMs), available from Github and CRAN.
Authors (Most recent CRAN version 4.2.2): Chun E., Broll S., Buchanan D., Muschelli J., Fernandes N. and others; **Gaynanova I.** (maintainer)
17. **ihr**: an R package for interpreting heart rate data from wearable devices, available from Github.
Authors: Yoo O., Han S. and **Gaynanova I.**

16. [SmoothHOOI](#): an R package for fitting smooth Tucker decomposition, available from Github.
Authors: [Qian L.](#), **Gaynanova I.**
15. [GlucoseBench](#): a collection of public CGM datasets with standardized pre-processing and common ML glucose forecasting models for glucose prediction, Python-based, available from Github.
Authors: [Sergazinov R.](#), [Chun E.](#), [Rogovchenko V.](#), [Fernandes N.](#), [Kasman N.](#) and **Gaynanova I.**
14. [fastfrechet](#): an R package for fast implementation of Frechet regression with distributional responses, available from GitHub.
Authors: [Coulter A.](#) (maintainer), [Lee R.](#) and **Gaynanova I.**
13. [singR](#): an R package with the implementation of SING method (SIMultaneous Non-Gaussian component analysis) for data integration in neuroimaging
Authors: [Wang L.](#), **Gaynanova I.**, [Risk B.](#)
12. [bp](#): an R package for blood pressure analysis, available from Github
Authors: [Schwenck, J.](#), **Gaynanova I.**
11. [latentcor](#): an R package for estimating latent correlations from mixed data types, available from Github
Authors: [Huang M.](#), [Yoon G.](#), [Müller, C.](#), **Gaynanova I.**
10. [SLIDE](#): an R package for learning partially-shared structures from multi-view data, available from Github.
Authors: **Gaynanova I.**, [Yuan D.](#)
9. [biClassify](#): an R package for binary classification using extensions of discriminant analysis, [CRAN](#)
Authors: [Lapanowski A.](#), **Gaynanova I.**
8. [Martin M.](#), [Buchanan D.](#), [Chun E.](#), [Bhat R.](#), [Cass S.](#), [Wang E.](#), [Senthil S.](#) and **Gaynanova I.** [✉] (2021). [irinagain/Awesome-CGM: Updated release with simulators and enhanced processing \(Version v1.1.0\)](#). Zenodo. DOI: [10.5281/zenodo.4723654](#)
7. [SPRING](#): an R package for estimation of sparse microbial association networks using rank-based correlation, available from Github.
Authors: [Yoon G.](#), **Gaynanova I.**, [Müller, C.](#)
6. [sparseKOS](#): an R package for nonlinear binary classification using sparse kernel optimal scoring, available from Github.
Authors: [Lapanowski A.](#), **Gaynanova I.**
5. [JACA](#): an R package for joint association and classification analysis of multi-view data, available from Github.
Authors: [Zhang Y.](#), **Gaynanova I.**
4. [mixedCCA](#): an R package for semiparametric sparse canonical correlation analysis for data of mixed types (continuous/ binary/ zero-inflated), available from Github.
Authors: [Yoon G.](#), **Gaynanova I.**
3. [DAP](#): an R package to perform discriminant analysis via projections, available from Github and CRAN.
Authors: [Wang T.](#), **Gaynanova I.**
2. [TRES](#): a Matlab package to perform sparse linear regression using TRES, available from Github.
Authors: [Müller C.](#), [Bien J.](#), **Gaynanova I.**, [Combettes P.](#)
1. [MGSDA](#): an R package to perform sparse multi-group discriminant analysis, available from CRAN.
Authors: **Gaynanova I.**

Funding Support

Active awards

- Bold Challenges Pollination Award** **Gaynanova I (PI)** **2026-2027**
 Planning Workshop for the Algorithmic and Statistical Foundations of Chronic Illness Management at a Distance with mHealth Data
 Total awarded amount: \$10,000.00
- Laude Moonshot Program** **Wiens J (PI)** **2026**
 An AI-enabled Conversational Advisor for Individualized Glucose Management
 Role: Co-PI
 Total awarded amount: \$100,000.00
- Michigan-Israel Partnership** **Gaynanova I (PI), Segal E (PI Weizmann)** **2026 - 2028**
 Predictive modeling using CGM and Multimodal Data for Personalized Management and Prevention of Diabetes and Metabolic Dysfunction
 Total awarded amount: \$100,000.00
- NIH R01 DK126373** **de Boer, I (PI, University of Washington)** **09/2025 - 08/2030**
 Blood Sugar Sensing on Maintenance Dialysis
 Role: Co-Investigator, 10% effort (single PI for UM)
 Amount attributed to UM: \$414,941.00
- NIH R01HL172785** **Gaynanova I. (PI)** **06/2024 - 02/2029**
 New machine learning methods for extracting features from digital health data with applications to sleep apnea
 Total awarded amount: \$1,780,503.00
- NSF CAREER Award DMS-2422478 (2044823 at TAMU)** **Gaynanova I. (PI)** **06/2021 - 05/2026**
 Next-generation methods for statistical integration of high-dimensional disparate data sources
 Total awarded amount: \$400,000.00
- NIH U24 DK122927** **Spino C. (PI)** **03/2023 - 12/2027**
 University of Michigan Data Coordinating Center for the Diabetic Foot Consortium
 Role: Co-investigator, 10% effort starting 01/2024

Past awards:

- TAMU Seed Grant for Promoting Research Collaborations** **Gaynanova I. (PI)** **06/2022 - 05/2023**
 Modifiable risk factors and interventions in diabetes
 Total awarded amount: \$10,000.00
- TAMU Presidential Transformation Teaching Grant (PTTG)** **Gaynanova I. (PI)** **01/2021 - 12/2022**
 Total awarded amount: \$20,000.00

- TAMU College of Science **Gaynanova I. (PI)** **10/2021 - 06/2022**
 Science Undergraduate Research Opportunities Program (SUROP)
 Total awarded amount: \$1,820.00 (support for 1 undergraduate student).
- TAMU Institute of Data Science **Ni Y. (PI), Gaynanova I. (Co-PI)** **07/2020 - 06/2022**
 Studying microbial interactions and host heterogeneity via data integration
 Total awarded amount: \$59,100.00 (50% of postdoctoral researcher support for 2 years)
- NSF CCF-1934904 **Mallick B. (PI)** **10/2019 - 09/2022**
 HDR Tripods: Texas A&M Research Institute for Interdisciplinary Foundations of Data Science
 Role: Co-Investigator, 0% effort (postdoctoral researcher support for 2 years)
- Johns Hopkins University, Subcontract **Gaynanova I. (PI)** **07/2019 - 06/2020**
 Statistical Analysis of CGM Data
 Total awarded amount: \$30,063.00
- NSF DMS-1712943 **Gaynanova I. (PI)** **07/2017 - 06/2020**
 Scalable methods for classification of high-dimensional heterogeneous data
 Total awarded amount: \$162,539.00

Presentations

Invited Presentations:

93. Northwestern University, Biostatistics and Informatics seminar, Chicago, IL, May 2026
92. Department of Pediatric Endocrinology, University of Michigan, Ann Arbor, MI, March 2026
91. ENAR, Indianapolis, IN, March 2026
90. ASA STATBOLIC conference, Rockville, MD, February 2026
89. MeTRIC Symposium, Ann Arbor, MI, January 2026
88. 2025 IMS International Conference on Statistics and Data Science, Seville, Spain, December 2025
87. DIGIT-MI session, Eisenberg Family Depression Center, University of Michigan, virtual, November 2025
86. Diabetic Foot Consortium Steering Committee Meeting, Tucson, AZ, November 2025
85. Invited Roundtable Facilitator, Caswell Diabetes Institute Metabolism, Obesity, Nutrition & Diabetes Symposium, Ann Arbor, MI, October 2025
84. International Day of Women in Statistics and Data Science, virtual, October 2025
83. International Statistical Institute World Statistics Congress, The Hague, Netherlands, October 2025
82. Department of Statistics, University of Michigan, September 2025
81. Invited presentation at Takeda Quantitative Sciences Digital team, virtual, August 2025
80. Invited panelist, session organized by Committee on Applied and Theoretical Statistics of the National Academies of Science, Engineering and Medicine, Joint Statistical Meetings, Nashville, TN, August 2025

79. Big Data Summer Institute (BDSI), Ann Arbor, MI, July 2025
78. TRICAP'25 Tensor and multiblock data analysis, Alesund, Norway, June 2025
77. Americal Statistical Association (ASA), section on Statistical Learning and Data Science (SLDS) webinar, virtual, April 2025
76. Test journal invited paper webinar, invited discussant, virtual, March 2025
75. ENAR, New Orleans, March 2025
74. BIRS (Banff International Research Station) Workshop on Emerging Statistical Methods for Digital Health Data, Banff, Canada, February 2025
73. Applied Biostatistical Sciences (ABS) Network lecture series, virtual, December 2024
72. NIH NHLBI virtual workshop on Heterogeneity and Sex/Gender Differences in Obstructive Sleep Apnea : Personalized Approaches to Diagnosis, Treatment, and Implementation, November 2024
71. Statistical Learning and Data Science Conference, Newport Beach, CA, November 2024
70. Roundtable Facilitator, Caswell Diabetes Institute Metabolism, Obesity, Nutrition & Diabetes Symposium, Ann Arbor, MI, October 2024
69. Invited session, Joint Statistical Meetings, Portland, OR, August 2024
68. The Big Data Summer Institute's (BDSI) Concluding Symposium, Ann Arbor, MI, July 2024
67. U.S. Food and Drug Administration, Center for Drug Evaluation and Research, April 2024
66. ENAR, Baltimore, MD, March 2024
65. Pre-ENAR Digital Health Methods Workshop, Baltimore, MD, March 2024
64. Department of Statistics and Probability, Michigan State University, September 2023
63. Celebration of Statistics and Data Science, Cornell University, September 2023
62. Department of Data Science and Operations, University of Southern California Marshall School of Business, March 2023
61. Department of Biostatistics, University of Washington, November 2022
60. Invited session, Joint Statistical Meetings, Washington DC, August 2022
59. TRICAP'22 Tensor and multiblock data analysis, Auberge Des Sequins, Luberon, France, June 2022
58. Department of Mathematics and Statistics, University of Massachusetts Amherst, May 2022
57. ENAR, Houston, TX, March 2022
56. Center for Approximation and Mathematical Data Analytics (CAMDA) seminar, Texas A&M University, March 2022
55. Department of Biostatistics, Virginia Commonwealth University, virtual due to COVID-19, December 2021
54. Missing values and Causal Inference seminar, joint between Ecole Polytechnique and INRIA Saclay, France, virtual due to COVID-19, November 2021
53. TAMIDS workshop on UQ and machine learning, Texas A&M University, November 2021
52. Department of Statistics, Penn State University, September 2021
51. ICSA Applied Statistics Symposium, virtual due to COVID-19, September 2021
50. Department of Statistics, University of Washington, virtual due to COVID-19, May 2021

49. Data & Science podcast with Glen Wright Colopy, April 2021
48. Transformational Teaching and Learning Conference, Texas A&M University, virtual due to COVID-19, April 2021
47. Pod of Asclepius, The Healthcare Technology Podcast sponsored by ASA, December 2020
46. Department of Mathematics, Statistics and Computer Science, the University of Illinois at Chicago, virtual due to COVID-19, October 2020
45. Department of Biostatistics, University of North Carolina, Chapel Hill, virtual due to COVID-19, August 2020
44. Invited session, Joint Statistical Meetings, virtual due to COVID-19, August 2020
43. 4th International Conference on Econometrics and Statistics (EcoSta 2020), Seoul, South Korea, July 2020 **(canceled due to COVID-19)**
42. Department of Biostatistics, University of Washington, May 2020 **(canceled due to COVID-19)**
41. Department of Statistics, University of Washington, May 2020 **(canceled due to COVID-19)**
40. Department of Biostatistics, Emory University, April 2020 **(canceled due to COVID-19)**
39. Banff International Research Station, workshop on the "Use of Wearable and Implantable Devices in Health Research", Banff, Canada, February 2020
38. Department of Statistics, Oregon State University, February 2020
37. The Gatsby Computational Neuroscience Unit, University College London, UK, December 2019
36. The Fourth Workshop on Higher-Order Asymptotics and Post-Selection Inference (WHOA-PSI), St. Louis, MO, August 2019
35. Topic-Contributed session, Joint Statistical Meetings, Denver, CO, July 2019
34. ICSA Applied Statistics Symposium, Raleigh, NC, June 2019
33. Econometrics and Statistics Colloquium, Booth School of Business, University of Chicago, May 2019
32. Department of Biostatistics, University of Michigan, April 2019
31. Big Data Working group, College of Veterinary Medicine, Texas A&M University, March 2019
30. Friday Science Seminar, College of Science, Texas A&M University, February 2019
29. Department of Biostatistics, University of Minnesota, February 2019
28. 11th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM Statistics 2018), Pisa, Italy, December 2018
27. Bioinformatics and Cancer Symposium, Texas A&M University, September 2018
26. Topic-Contributed session, Joint Statistical Meetings, Vancouver, Canada, August 2018
25. IMS Asia Pacific Rim Meeting 2018, Singapore, June 2018
24. ICSA Applied Statistics Symposium, New Brunswick, NJ, June 2018
23. Statistical Learning and Data Science Conference, New York, NY, June 2018
22. Department of Statistics, Indiana University, April 2018
21. SAMSI Operator Splitting Workshop, Research Triangle Park, NC, March 2018
20. Department of Biostatistics, UT MD Anderson Cancer Center, March 2018

19. 10th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM Statistics 2017), London, UK, December 2017
18. Department of Statistical Science, Baylor University, November 2017
17. Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, September 2017
16. Data-Driven Model Reduction Workshop, Texas A&M University, April 2017
15. Conference of Texas Statisticians (COTS), Dallas, TX, March 2017
14. ENAR, Washington DC, March 2017
13. 9th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM Statistics 2016), Seville, Spain, December 2016
12. Bio-seminar, Department of Electrical and Computer Engineering, Texas A&M University, October 2016
11. Topic-Contributed session, Joint Statistical Meetings, Chicago, IL, August 2016
10. Southern Regional Council on Statistics Summer Research Conference, Bentonville, AR, June 2016
9. Structured Multivariate Data Workshop, Texas A&M University, January 2016
8. Department of Statistics, Rice University, February 2015
7. Department of Statistical Science, Cornell University, February 2015
6. Department of Statistics, Indiana University, January 2015
5. Department of Statistics, Texas A&M University, January 2015
4. Department of Mathematical and Statistical Sciences, University of Colorado, Denver, January 2015
3. Department of Statistics and Actuarial Science, University of Iowa, January 2015
2. Department of Statistics, University of California, Davis, January 2015
1. Department of Biostatistics, University of Iowa, December 2014

Contributed Presentations:

14. Poster, R/Medicine 2021, virtual due to COVID-19, August 2021
13. Presentation, ENAR, virtual due to COVID-19, March 2021
12. Poster, ENAR, Philadelphia, PA, March 2019
11. IMS New Researchers Conference, Baltimore, MD, August 2017
10. Poster, International Conference on Machine Learning in New York City, NY, June 2016
9. Contributed session, Joint Statistical Meetings in Boston, MA, August 2014
8. Statistics Student Seminar, Cornell University, April 2014
7. Poster, SAMSI LDHD Workshop, February 2014
6. Contributed session, Joint Statistical Meetings in Montreal, Canada, August 2013
5. Poster, SAMSI LDHD Summer School, August 2013
4. Statistics Student Seminar, Cornell University, March 2013
3. Contributed session, Joint Statistical Meetings in San Diego, CA, August 2012
2. Cross-Campus Collaborative Colloquium, Cornell University, December 2011
1. Statistics Student Seminar, Cornell University, September 2011

Teaching Experience

Primary Instructor (at University of Michigan):

BIOS 601: Probability and Distribution Theory

Fall 2025

Graduate-level introduction to probability and distribution theory needed for statistical inference. Probability, discrete and continuous distributions, expectation, generating functions, limit theorems, transformations, sampling theory.

BIOS 699: Analysis of Biostatistical Investigations

Winter 2024/2025

Identifying and solving design and data analysis problems using a wide range of biostatistical methods. Written and oral reports on intermediate and final results of case studies required.

Table 1: Summary of teaching evaluations at University of Michigan, average of the median response to three university-wide questions about the instructor (class preparation, clear explanation of material, treating students with respect). Scale 1(Strongly Disagree) to 5(Strongly Agree).

Course	Semester	Total class size	Mean response
BIOS 699	Winter 2024	87	4.6
BIOS 699	Winter 2025	71	4.8
BIOS 601	Fall 2025	61	4.9

Primary Instructor (at Texas A&M University):

STAT 211: Principles of Statistics I

Fall 2015/2016/2017, Spring 2016

Calculus-based introduction to probability and probability distributions; sampling and descriptive measures; inference and hypothesis testing; analysis of variance; linear regression.

STAT 489: Structured Research Experiences in Statistics

Spring 2022

Research project based on simulation studies in statistics; version control; reproducible computations; simulation study design; publication-quality figures; scientific writing.

STAT 610: Distribution Theory

Fall 2017/2018/2019/2020/2021/2022

Graduate-level introduction to probability theory; distributions and expectations of random variables, transformations of random variables and order statistics; generating functions and basic limit concepts.

STAT 689/STAT 695: Statistical Learning with Sparsity

Spring 2017/2020

Graduate-level class covering penalized empirical loss minimization methods with sparsity-inducing penalties. The course also includes brief introduction to convex optimization and duality.

STAT 600: Computational Statistics

Spring 2018, Fall 2019/2020/2021/2022

Graduate-level course on computational statistics and optimization. Topics include version control with Git and Github, code vectorization and profiling, writing R packages, introduction to convex optimization and optimization algorithms.

STAT 618: Statistical aspects of Machine Learning II

Spring 2023

Table 2: Summary of teaching evaluations at Texas A&M University, Q: On the whole, this was a good instructor. Scale 1(Strongly Disagree) to 5(Strongly Agree).

Course	Semester	Total class size	Mean response
STAT 211	Fall 2015	77	4.7
STAT 211	Spring 2016	77	4.2
STAT 211	Fall 2016	97	4.6
STAT 211	Fall 2017	97	4.4
STAT 489	Spring 2022	8	5.0
STAT 610	Fall 2017	35	4.8
STAT 610	Fall 2018	34	5.0
STAT 610	Fall 2019	39	4.5
STAT 610	Fall 2020	33	4.8
STAT 610	Fall 2021	34	4.9
STAT 610	Fall 2022	45	4.6
STAT 600	Spring 2019	7	4.8
STAT 600	Fall 2019	31	4.7
STAT 600	Fall 2020	12	4.9
STAT 600	Fall 2021	20	5.0
STAT 600	Fall 2022	16	4.9
STAT 618	Spring 2023	8	5.0
STAT 689	Spring 2017	12	5.0
STAT 695	Spring 2020	15	5.0

Graduate-level course on statistical machine learning, a continuation of STAT 616. Topics include advanced material on statistical learning with sparsity, low-rank matrix decompositions, tensor methods, deep learning, and non-euclidean machine learning.

Primary Instructor (at Cornell University):

ILRST 2100: Introductory Statistics

Winter 2013, 2014 ,2015

Non-calculus-based introduction to statistics; sampling and descriptive measures; inference and hypothesis testing; analysis of variance; linear regression.

Teaching Assistant (at Cornell University):

ILRST 2100: Introductory Statistics

Summer 2013, 2014

Non-calculus-based introduction to statistics; sampling and descriptive measures; inference and hypothesis testing; analysis of variance; linear regression.

ENGRD 2700: Basic Engineering Probability and Statistics (recitation leader)

Fall 2013

Calculus-based introduction to probability and probability distributions; sampling and descriptive measures; inference and hypothesis testing; analysis of variance; linear regression.

MATH 4720: Statistics

Spring 2015

Introduction to mathematical statistics based in calculus, linear algebra, and probability theory; inference for proportions, inference for means, contingency tables and linear regression.

BTRY 6010: Statistical Methods I (TA/Lab instructor)

Fall 2009, 2010, 2014

Graduate level introduction to statistical methods for analyzing data; descriptive statistics and data visualization; analysis of variance; linear regression.

BTRY 6020: Statistical Methods II (TA/Lab instructor)

Spring 2010, 2011, 2012

Continuation of BTRY 6010. Emphasizes the use of multiple regression analysis, experimental design and generalized linear models.

Mentoring

Postdoctoral Researchers

- | | |
|---|--------------------------|
| 1. Junyoung Park | 09/2024 - present |
| 2. Sangyoon Yi (with Raymond Wong) | 09/2020 - 08/2022 |
| HDR Tripods: Texas A&M Research Institute for Interdisciplinary Foundations of Data Science | |
| 3. Hee Cheol Chung (with Yang Ni) | 06/2020 - 06/2022 |
| TAMU Institute of Data Science Postdoctoral Project Program | |
| 4. Grace Yoon | 08/2017 - 08/2020 |
| Postdoctoral Trainee of DHHS-NIH National Cancer Institute T32CA090301 (PI: R. Carroll), Gaynanova is the Statistics Mentor | |

PhD Committee Chair

At University of Michigan:

- | | |
|--|-----------------------|
| 1. Leyuan Qian | 2028(expected) |
| 2. Edward Shao (co-chair with Hui Jiang) | 2027(expected) |

At Texas A&M University:

- | | |
|---|-------------|
| 3. Alexander Coulter (co-chair with Yuchao Jiang) | 2026 |
| 4. Lei Wang (co-chair with Jade Wang) | 2026 |
| 5. Renat Sergazinov | 2024 |
| 6. Mingze Huang (ECON) | 2022 |
| 7. Dongbang Yuan | 2022 |
| 8. Alex Lapanowski | 2020 |
| 9. Yunfeng Zhang | 2020 |
| 10. Tianying Wang (co-chair with Raymond Carroll) | 2018 |

PhD Committee Member

At University of Michigan:

- | | |
|---|------------------------|
| 1. Benjamin Kessler (chair Sung Kyun Park, Major: Epidemiology) | 2028 (expected) |
|---|------------------------|

- | | |
|---|-----------------|
| 2. JungMin Lee (chair Jenna Wiens, Major: Computer Science and Engineering) | 2026 (expected) |
| 3. M Crystal Wen (chair Josh Welch, Major: Bioinformatics) | 2027 (expected) |
| 4. Mukai Wang (chair Gen Li, Major: Biostatistics) | 2026 (expected) |
| 5. Qingzhi Liu (chairs Veera Baladandayuthapani and Gen Li, Major: Biostatistics) | 2025 |

At Texas A&M University:

- | | |
|---|-----------------|
| 6. Carson James (chair Jesus Arroyo, Major: Statistics) | 2026 (expected) |
| 7. Connor Brubacker (chair Scott Bruce, Major: Statistics) | 2025 |
| 8. Jian Yan (chair Xianyang Zhang, Major: Statistics) | 2023 |
| 9. Junsouk Choi (chair Yang Ni, Major: Statistic) | 2023 |
| 10. Krystin Pantoja (chair David Jones, Major: Statistics) | 2022 |
| 11. Jiayi Wang (chair Raymond Wong, Major: Statistics) | 2022 |
| 12. Jianling Wang (chair James Caverlee, Major: Computer Science and Engineering) | 2022 |
| 13. Nida Obatake (chair Anne Shiu, Major: Mathematics) | 2021 |
| 14. Allyson Souris (chair Anirban Bhattacharya, Major: Statistics) | 2020 |
| 15. Sangyoon Yi (chair Xianyang Zhang, Major: Statistics) | 2020 |

MS Committee Chair*At Texas A&M University:*

1. Mingze Huang	2022
2. John Schwenck	2021
3. Rucha Bhat	2021
4. Shaun Cass	2021
5. Matthew Goldsmith	2021
6. Neil Kirkpatrick	2021

MS Committee Member*At Texas A&M University:*

1. Eric Riley (STAT)	2019
2. Andrea Barton (MATH)	2023
3. Polina Churilova (PETE)	2023
4. Aaron Knodell (CPSC)	2019
5. Haley Pichler (MATH)	2019
6. Jung Sim Hyun (MATH)	2019

Graduate Student Research Assistantship (GSRA)*At University of Michigan:*

1. Weixiong Hua	Winter 2026 - present
2. Nathan Szeto (through DFC, PI Spino C.)	Fall 2025 - present
3. Leyuan Qian, PhD, Biostatistics	Fall 2024 - present
4. Edward Shao, PhD, Biostatistics	Summer 2024/Fall 2024/Winter 2025/Summer 2025
5. Neo Kok, MS, Health Data Science	Fall 2024 - May 2026
6. Walter Williamson, MS, Biostatistics	Winter 2025

Other Graduate Student Research*At University of Michigan:*

1. Catherine Yan	Winter 2026 - present
2. Sylvan Xu, MS, Biostatistics (with Gen Li, Sunway Trust scholarship Summer 24)	Summer 2024/Fall 24/Winter 2025
3. Walter Williamson, MS, Biostatistics (IHPI scholarship Summer 24)	Summer 2024/Fall 24/Winter 2025
4. Charlotte Xu, MS, Biostatistics	Winter 2024/Fall 2024/Winter 2025

Undergraduate Student Research*At University of Michigan:*

1. Rachel Shi (STAT)	Winter 2026
2. Zhihui Chen (STAT)	Winter 2026

- | | |
|-------------------------------------|-----------------------------------|
| 3. Owen Yoo (STAT), Honors Thesis | Winter 2025/Fall 2025/Winter 2026 |
| 4. Sophia Han (STAT) | Winter 2025 |
| 5. Samuel Tan (STAT), Honors Thesis | Fall 2024/Winter 2025 |

At Texas A&M University:

- | | |
|---|-------------------------------------|
| 6. Nathaniel Fernandes (ENGE) | 08/2020 - 08/2021; Spring 2023 |
| 7. Akhil Cutinha (CPSC) | Fall 2022 |
| 8. Urjeet Shrestha (APMS) | Fall 2022 |
| 9. Nicholas Kasman (ENGE) | Spring 2022, Fall 2022, Spring 2023 |
| 10. Elizabeth Chun (BMCB/STAT) | 01/2020 - 05/2023 |
| 11. Melinda Chen (STAT), joint mentoring with Jesus Arroyo | Summer 2022 |
| 12. Wesley Halstead (STAT), joint mentoring with Jesus Arroyo | Summer 2022 |
| 13. Sina Mokhtar (STAT) | Summer 2022 |
| 14. Valerie Espinosa (STAT) | Summer 2022 |
| 15. Devon Maywald (APMS) | 06/2021 - 08/2022 |
| 16. Yixin Zhang (STAT) | Summer 2021 |
| 17. Kyle Schichl (STAT) | Summer 2021 |
| 18. Zengxiaoran Kang (STAT) | Summer 2021 |
| 19. Benjamin Adams (STAT) | Summer 2021 |
| 20. Jada Lawson (STAT) | Summer 2021 |
| 21. Mayra Hernandez (STAT) | Summer 2021 |
| 22. Johnathan Shih (STAT) | Spring 2021 |
| 23. Jung Hoon Seo (CPSC) | Spring 2021 |
| 24. Marielle Hicban (BMEN) | 06/2020 - 05/2021 |
| 25. David Buchanan (STAT) | 01/2020 - 05/2021 |
| 26. Ashok Meyyappan (SPSC) | Fall 2020 |
| 27. Nhan Nguyen (ENGE) | 06/2020 - 12/2020 |
| 28. Pratik Patel (STAT) | 06/2020 - 12/2020 |
| 29. Mary Martin (STAT) | 01/2020 - 12/2020 |
| 30. Sangaman Senthil (INEN) | Spring 2020 |
| 31. Eric Wang (ENGE) | Spring 2020 |
| 32. Steven Broll (STAT) | 06/2019 - 08/2020 |

Academic advising*At University of Michigan:*

- Yize Hao (PhD), Neo Kok (MS), Nandhana Nambi (MS), Leyuan Qian (PhD), Edward Shao (PhD), Haoran Tang (PhD), Xinyuan Zhang (MS), Samuel Tan (MS), Medha Rao (MS) **2025/2026**
- Zhuoxin Fu (MS), Yize Hao (PhD), Neo Kok (MS), Anqi Li (MS), Nandhana Nambi (MS), Leyuan Qian (PhD), Medha Rao (MS), Haoran Tang (MS), Xinyuan Zhang (MS) **2024/2025**
- Zhuoxin Fu (MS), Anqi Li (MS), Jianxiong Chen (MS), Haoran Tang (MS) **2023/2024**

Other formal mentoring

At the University of Michigan:

- Mentorship committee for Jean Morrison, Assistant Professor in Biostatistics 2024-2026
- Mentorship committee for Walter Dempsey, Assistant Professor in Biostatistics 2023-2024

At Texas A&M University:

- Mentorship committee for Nilanjana Laha and Jesus Arroyo, Assistant Professors in Statistics 2022-2023
 - Brittany Segundo 01/2019 - 05/2023
- Dr. Gaynanova is a teaching mentor through the Academy for Future Faculty (AFF)

Departmental and University Service

At University of Michigan:

Department of Biostatistics Committees:

- Student Awards Committee 2025/2026
- Retreat Planning & 2030 Visioning Committee 2024/2025
- Faculty Search Committee 2024/2025
- Candidacy Committee 2023/2024
- Diversity, Equity and Inclusion Committee 2023/2024
- Biostatistics Faculty Retreat 2023 working groups on AI in Research and New student cohort 2023

School of Public Health Committees:

- SPH Executive Committee 2025 - present
- M-PACT Search Committee 2024/2025

Other:

- Reviewer for Student Abstract Competition, IDEAS - Combating Infectious Diseases initiative 2023

At Texas A&M University:

Department of Statistics Committees:

- Faculty Advisory Committee 2022/2023
- Promotion & Tenure Committee (Elected member) 01/2022 - 08/2023
- SOAR subcommittees (Faculty workgroup, Research Bio workgroup) 2021
- Graduate Committee 2019/2020; 2020/2021; 2021/2022; 2022/2023
- Faculty Evaluation and Promotion Guidelines Committee 2020
- Hiring Committee 2018/2019; 2020/2021
- Colloquium Chair 05/2018 - 12/2019
- Undergraduate Committee 2017/2018

Panelist:

- TAMU Workshop on NSF CAREER and other Young Investigator awards 11/2021

- 3rd Annual TX-LA Undergraduate Mathematics conference 10/2019
- TAMU College of Science, Lunch and Learn series 09/2019
- TAMU Symposium for Faculty, Staff, Graduate Students and PostDocs in the Sciences 02/2017
- TAMU AWM Chapter Panel "Career in Mathematics" 03/2016

Other:

- TASTE (Take a Scientist to Lunch) College of Science Program 04/2022
- Faculty advisor for Statistical Learning Journal Club, Department of Statistics 2019 - 2020
- Friday Science Seminar, College of Science, Texas A&M University 02/2019

At Cornell University:

- President, STATS graduate student organization 11/2013 - 05/2015
- Organiser, Statistics Student Seminars 08/2011 - 12/2013

Professional Activities

Editorial service:

- Associate Editor, *Annals of Applied Statistics* 2024 - present
- Associate Editor, *Biometrika* 2023 - present
- Associate Editor, *Journal of the American Statistical Association - Theory & Methods* 2023 - present
- Associate Editor, *Data Science in Science* 2021 - present
- Associate Editor, *Journal of Computational and Graphical Statistics* 2018 - 2022

Journal refereeing:

- Annals of Applied Statistics
- Bernoulli
- Bioinformatics
- Biometrics
- Biometrika
- Computational Statistics
- Communications Medicine
- Data Science in Science
- Diabetes Technology and Therapeutics
- Electronic Journal of Statistics
- Journal of the American Statistical Association
- Journal of Computational and Graphical Statistics
- Journal of Machine Learning Research
- Journal of Multivariate Analysis
- IEEE Transactions on Network Science and Engineering
- Nature Medicine
- PLOS One
- R journal
- Statistical Analysis and Data Mining

- Statistical Modeling: an International Journal
- SoftwareX
- Stat
- Statistics in Biosciences
- Statistics in Medicine
- Statistics & Probability Letters
- Statistica Sinica
- Technometrics
- TEST

Conference refereeing:

- NIPS (Neural Information Processing Systems) 2020
- AISTATS (Artificial Intelligence and Statistics) 2020

Book proposals refereeing:

- Springer 2018
- CRC/Chapman & Hall 2017, 2018
- Wiley 2017
- MacMillan Education 2015

Grant review panels:

- NIH Clinical Data Management and Analysis (CDMA) 2026
- NIH Special Emphasis Panel/Scientific Review Group 2025/10 HLBP (25) 2025
- 2024 Trans-Omics for Precision Medicine (TOPMed) Fellowship 2025
- PCORI Improving Methods for Conducting Patient-Centered Outcomes Research 2023
- NIH, Small Business: Computational, Modeling, and Biodata Management – MCST 2022
- NSF Division of Mathematical Sciences 2021, 2024
- Joint NSF/NIH panel 2016

External reviews:

- External reviewer for 1 case of promotion from Assistant to Associate Professor with tenure 2025
- External reviewer for 1 case of promotion from Assistant to Associate Professor with tenure 2023
- External reviewer for 1 case of promotion from Assistant to Associate Professor with tenure 2022

Other reviews:

- Invited reviewer for the The National Academies of Sciences, Engineering, and Medicine report on "Frontiers of Statistics in Science and Engineering: 2035 and Beyond (2026)" 2025

Elected Positions:

- Chair-Elect 2026, Chair 2027, ASA Interest Group on Mobile and Wearable Data Science 2026-2027

- Program Chair-Elect 2024, Program Chair 2025, ASA section on Statistical Learning and Data Science (SLDS) **2023-2025**
- President, Southeast Texas Chapter of ASA (SETCASA) **02/2019-01/2020**
Main activity: led, coordinated and judged SETCASA annual poster competition
- Vice-president, Southeast Texas Chapter of ASA (SETCASA) **10/2017-01/2019**
Main activity: co-organized and judged SETCASA annual poster competition

Professional Committees Chair:

- Chair of ASA SLDS Student Paper Award 2021 Committee **2020/2021**
- Co-chair (chair Genevera Allen) of ASA SLDS Student Paper Award 2020 Committee **2019/2020**

Professional Committees Member:

- ASA SLDS Breiman Award 2022 Committee **2021/2022**
- SLDS conference Student Paper Award Committee **2020**
- ASA SLDS JSM 2019 Poster Competition Committee **2019**
- ASA SLDS Student Paper Award 2019 Committee **2018/2019**
- COTS (Conference of Texas Statisticians) Poster Competition Committee **2017**
- ASA SCS John M. Chambers Statistical Software Award Committee **2015/2016**

Professional Panels:

- NISS Leadership Panel organizer and moderator, "Leadership in Translational Statistics and Machine Learning: Working Hand-in-Hand with the Clinical World" **03/2026**
- ASA CFR Webinar: Peer Advice for Statisticians: Writing Your First NSF Proposal **11/2023**
- NISS Fall 2021 Virtual Academic Career Fair; moderator **09/2021**
- NISS Fall 2020 Virtual Academic Career Fair; presenter **12/2020**

Conference Program Committees:

- ENAR 2027, Institute of Mathematical Statistics representative **2026-2027**
- Statistical Foundations of AI, Scientific Program Committee, Santa Barbara, CA **2026**
- JSM 2025, Statistical Learning and Data Science section representative (roundtable organization for JSM 2024) **2024 - 2025**
- Conference on Statistical Learning and Data Science (SLDS) 2024 **2024**
- Statistical Methods for High-Dimensional Complex Data, in honor of Dr. Raymond Carroll **2024**
- 13th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM Statistics 2020) in London, UK **2020**
- ENAR 2019, Statistical Learning and Data Science section representative **2019**

Sessions Organized:

- Invited Session at ENAR: "Statistical Innovations for Functional, Longitudinal, and Tensor Data from Wearable and Mobile Technologies" **03/2026**
- Invited Session at ENAR: "Quantile and Distributional Approaches in mHealth" **03/2025**

- Invited Session at SLDS 24 conference: "Statistical and machine learning methods for data from wearable and mobile devices" **11/2024**
- Invited Session at Joint Statistical Meetings: "New frontiers in data integration" **08/2024**
- Invited Session at CM Statistics in London, UK: "Statistical methods for time-varying multivariate data" **12/2019**
- Topic-Contributed Session at Joint Statistical Meetings: "Integrative approaches for statistical analysis of data from multiple sources" **08/2019**
- Invited Session at Joint Statistical Meetings: "Discovering homology in multi-view data: new statistical methods for data integration" **08/2018**
- Topic-Contributed Session at Joint Statistical Meetings: "Exploiting Low-Dimensional Structures: Recent Advances of Statistical Learning Methods in Genetics and Genomics" **08/2016**

Membership:

- American Statistical Association (ASA) **since 2011**
- Institute of Mathematical Statistics (IMS) **since 2012**
- ENAR **since 2015**